MEMORANDUM	**PROJECT: UniHive™ Closed-Loop Campus Marketplace & Regulated Dispatch Ecosystem**
TO:	The Faculty Academic Advisor / Senior Lecturer, Department of Computer Science & Directorate of Student Affairs, Redeemer's University (RUN), Ede, Osun State.
FROM:	Student Founder & Lead Systems Architect, Department of Computer Science, Redeemer's University.
SUBJECT:	**Proposal for Faculty Mentorship, Institutional Endorsement, and Technical Integration with RUN's Incumbent Campus Logistics Partner to Formalize Student Commerce & Generate University IGR.**

1. Executive Summary & Project Statement

UniHive™ is a specialized, institutional closed-loop e-commerce and secure logistics mediation platform engineered specifically for Nigerian private collegiate residential environments, with **Redeemer's University (RUN), Ede** positioned as the premier flagship launch campus. The software architecture directly addresses the acute risks of informal 'WhatsApp status' commerce — eliminating student prepayment fraud, eradicating hostel boundary infractions, and replacing unorganized trading with a formalized, digitally audited campus marketplace.

Crucially, **UniHive does not hire, vet, or manage physical delivery riders.** Instead, physical transit is exclusively contracted to the **licensed, accredited third-party logistics company already officially approved and operating inside Redeemer's University.** This strategic operational design relieves the platform and student founder of all courier vetting, security clearance, and transport liabilities, while providing the university's established logistics partner with optimized digital batch orders and guaranteed escrow settlement.

CAMPUS PILOT BASE	SECURITY ENGINE	LOGISTICS PARTNER	SCHOOL IGR SHARE
8,500+	**4-Digit PIN**	**Incumbent Co.**	**25% - 30%**
RUN Students & Residents	Porter Lodge Inspection Escrow	Already Operating in RUN	Automated Bursary Remittance

Core Strategic Objectives for Redeemer's University

1. Eradication of Campus WhatsApp Scams Transitions students from ephemeral WhatsApp status deals into an auditable marketplace protected by Paystack escrow and 4-digit PIN verification.	**2. Strict Preservation of Hostel Sanctity** All deliveries terminate strictly at the ground-floor porter lodge reception. Zero opposite-gender hallway access and zero student room intrusion.
3. Zero Logistics Vetting Burden Partnering directly with RUN's existing licensed campus logistics provider leverages their university gate clearance and pre-vetted personnel.	**4. Recurring University Internally Generated Revenue** Automated 25%–30% revenue share on all platform fees, yielding predictable monthly IGR for the University Bursary with NGN 0 capital expenditure.

2. The Problem Landscape at Redeemer's University (RUN)

Commercial activity among Redeemer's University students is vibrant, spanning fashion, thrift apparel, tech accessories, packaged snacks, and study materials. However, because no dedicated institutional infrastructure exists, over 85% of this commerce occurs across fragmented 24-hour WhatsApp statuses and informal group chats. This unregulated model creates severe institutional hazards:

RISK 1: PREPAYMENT FRAUD & GHOST MERCHANTS
Buyers transfer money upfront to unknown student accounts; vendors block buyers or fail to deliver. Fake bank SMS alerts and fabricated debit screenshots are rampant, leading to bitter student conflicts.

RISK 2: HOSTEL REGULATION BREACHES
Redeemer's University enforces strict gender boundaries across halls (Queen Esther, Deborah, Daniel, Joseph). Informal deliveries tempt couriers to sneak into hostel corridors, violating hall warden and proctorate rules.

RISK 3: COMMERCIAL DISPATCH DISRUPTION
Outside commercial bike riders cannot pass through security gates to reach student halls. Packages are frequently dumped at distant gates, resulting in lost parcels and security concerns.

RISK 4: ADMINISTRATIVE DISCIPLINARY OVERHEAD
The Directorate of Student Affairs and Campus Proctorate spend valuable hours arbitrating student fraud petitions, theft claims, and inter-student financial disputes that lack digital transaction records.

3. The UniHive Closed-Loop Solution Architecture

UniHive transforms this volatile landscape by introducing four interlocking institutional engineering pillars:

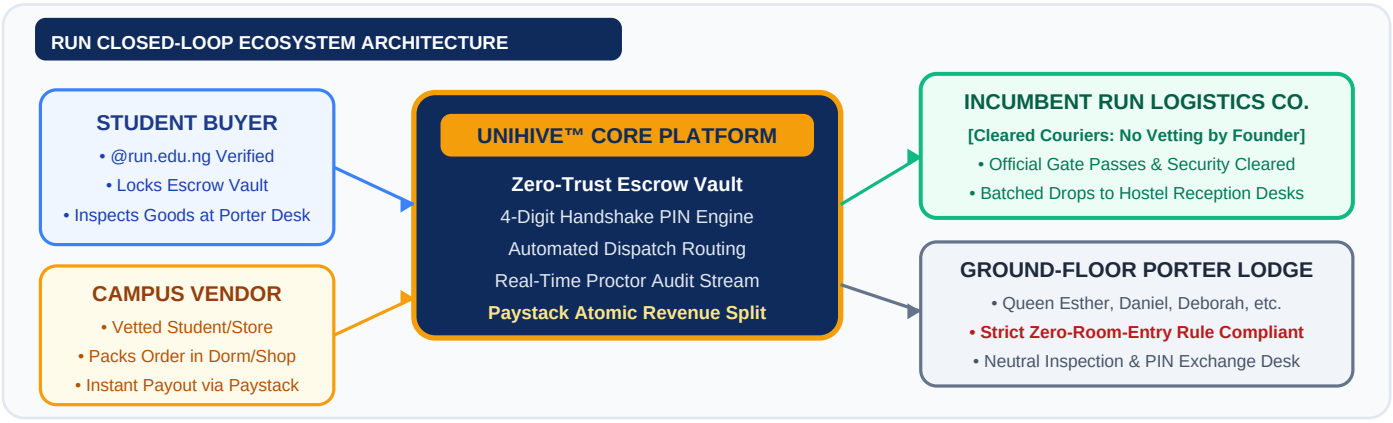
A. Institutional Identity Firewall (@run.edu.ng): Registration is restricted to students with authentic Redeemer's University email credentials and verified matriculation numbers. Outside anonymous fraudsters and unverified third parties are strictly excluded.

B. Zero-Trust Physical Inspection Escrow Engine: Payments made via Paystack are held in the UniHive Escrow Vault. Funds are never released upon checkout. The student buyer retains the right to physically inspect the merchandise (size, condition, functionality) before authorization.

C. The 4-Digit Handshake PIN Protocol: A unique, cryptographically hashed 4-digit PIN is generated upon order placement. Only when the buyer physically approves the item at the porter desk do they disclose this PIN to the courier. If 3 invalid attempts occur, the escrow automatically locks to prevent coercion.

D. The Ground-Floor Porter Lodge Policy: Couriers deliver strictly to the designated ground-floor porter lodge or security reception desk of RUN Halls of Residence (e.g., Queen Esther Hall Reception, Daniel Hall Security Desk). Couriers are strictly prohibited from residential corridors.

Figure 1: UniHive™ Redeemer's University Multi-Stakeholder Ecosystem Architecture



4. Strategic Logistics Partnership: Utilizing RUN's Incumbent Logistics Company

CRITICAL OPERATIONAL & LEGAL GUARANTEE:
The delivery of all UniHive orders will be handled exclusively by the licensed, accredited commercial logistics company already operating within Redeemer's University.
The platform founder and student software engineers **WILL NOT** recruit, vet, manage, or employ individual delivery riders. By integrating directly with RUN's incumbent logistics service provider, the entire burden of background screening, guarantor verification, physical security compliance, vehicle registration, and courier conduct remains with the already-established, university-cleared partner.

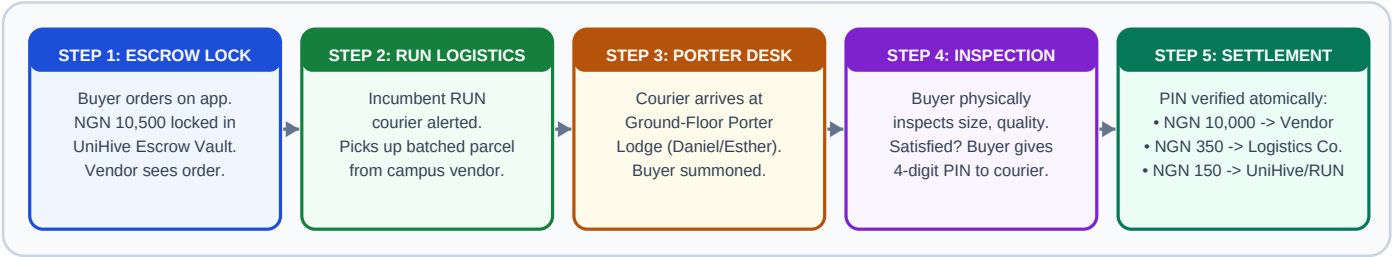
Why This Partner Model Is Strategically Superior for Redeemer's University:

1. Zero Vetting Overhead for Founder & School: Neither the student founder nor the university administration needs to establish a courier screening apparatus. The logistics company's couriers already hold legitimate campus passes and vetted proctorate status.	**2. Total Institutional Security Continuity:** No unauthorized or unknown outside delivery personnel are introduced onto RUN premises. Campus security checkpoints interact only with familiar, registered logistics staff.
3. Turnkey Digital Dispatch Terminal: UniHive provides the logistics company with an enterprise web & mobile dashboard. Incoming orders are automatically batched by destination hostel (e.g. 5 packages grouped for Queen Esther Hall).	**4. Guaranteed, Instant Cashflow for Logistics Partner:** Eliminates bad debts and delivery cash handling. Delivery fees (NGN 350 - NGN 500 per dropoff) are held in Paystack escrow and settled directly to the logistics company's bank account upon PIN confirmation.

The Step-by-Step Logistics & Escrow Handshake Sequence

The complete lifecycle from order placement to physical dropoff follows a rigorous 5-step sequence, verified via automated test suites in our codebase:

Figure 2: Zero-Trust Physical Inspection Escrow & 4-Digit Handshake PIN Sequence



Hostel Destination Topology (Pre-Seeded for Redeemer's University Pilot)

Hostel Name	Gender	Designated Delivery & Inspection Point	Sanctity Compliance
Queen Esther Hall	Female	Queen Esther Ground Floor Porter Lodge Reception	Strict Ground Floor Only (Zero Corridor Entry)
Deborah Hall	Female	Deborah Ground Floor Security Reception Desk	Strict Ground Floor Only (Zero Corridor Entry)
Dorcas Hall	Female	Dorcas Porter Reception Counter	Strict Ground Floor Only (Zero Corridor Entry)
Daniel Hall	Male	Daniel Security & Porter Desk	Strict Ground Floor Only (Zero Room Entry)
Joseph Hall	Male	Joseph Hall Central Porter Lounge Desk	Strict Ground Floor Only (Zero Room Entry)
Samuel Hall	Male	Samuel Hall Main Security Checkpoint	Strict Ground Floor Only (Zero Room Entry)

5. The Comprehensive Work & Role of the Administrator

A central requirement of university authorities is maintaining complete institutional oversight, moral compliance, and audit transparency. The UniHive **Admin Operations Command Center** (co-managed by University Representatives—such as the Directorate of Student Affairs, Campus Security Proctorate, or RUCED—and Platform Technical Operations) provides exhaustive administrative controls across 5 core pillars:

PILLAR 1: VENDOR ONBOARDING & CATALOG SCREENING	• Verifies matriculation numbers and bank subaccounts for student merchants. • Enforces strict institutional moral compliance: automated keyword filters and manual screening immediately block contraband, illicit substances, unprescribed pharmaceuticals, exam malpractice materials, alcohol, or weapons. • 1-click administrative delisting or store ban for non-compliant sellers.
PILLAR 2: REAL-TIME DISPATCH & TRANSIT OVERSIGHT	• Live operational stream tracking every package in transit across RUN campus. • Precise digital timestamps for: vendor pickup, logistics dispatch, arrival at porter lodge, and buyer handshake. • Provides campus proctors with immediate digital evidence if an item's origin or transit is ever queried.
PILLAR 3: ESCROW VAULT & FINANCIAL RECONCILIATION	• Real-time audit clean ledger tracking 100% of funds held in the Paystack escrow vault. • Monitors payout logs ensuring atomic settlement: zero fund leakage and zero untracked cash. • Generates comprehensive exportable CSV financial statements for University Bursary auditing.
PILLAR 4: ZERO-TRUST DISPUTE MEDIATION TERMINAL	• Manages automatic dispute escalations triggered by 3-strike PIN lockouts or buyer rejections. • Admin evaluates photo evidence, courier timestamps, and porter desk confirmation. • **1-Click Verdict Execution:** Either issues an immediate 100% Paystack refund to the buyer or releases legitimate funds to the vendor, extinguishing fraud complaints.
PILLAR 5: HARDENED ACCESS SECURITY (RFC 6238 2FA)	• Admin accounts are strictly pre-provisioned by university IT (no public registration). • Every administrative login requires **RFC 6238 TOTP Two-Factor Authentication** (Google/Microsoft Authenticator). • Prevents unauthorized access or credential theft, ensuring institutional data integrity.

The Admin Dispute Resolution & Anti-Fraud Lifecycle

When an irregular event occurs (such as a buyer claiming the wrong item arrived, or a runner failing to deliver), the escrow vault instantly locks and routes the case to the Admin Terminal for rapid administrative resolution:

Figure 3: Administrative Incident Handling & 1-Click Dispute Mediation Workflow



6. Revenue Model, Unit Economics & Financial Projections

The economic architecture of UniHive is engineered as a high-margin, zero-capital-expenditure utility that generates significant profitability for the student founder while delivering direct, hands-free **Internally Generated Revenue (IGR)** to Redeemer's University. No funds are requested from the university budget; rather, the platform generates institutional revenue from existing commerce.

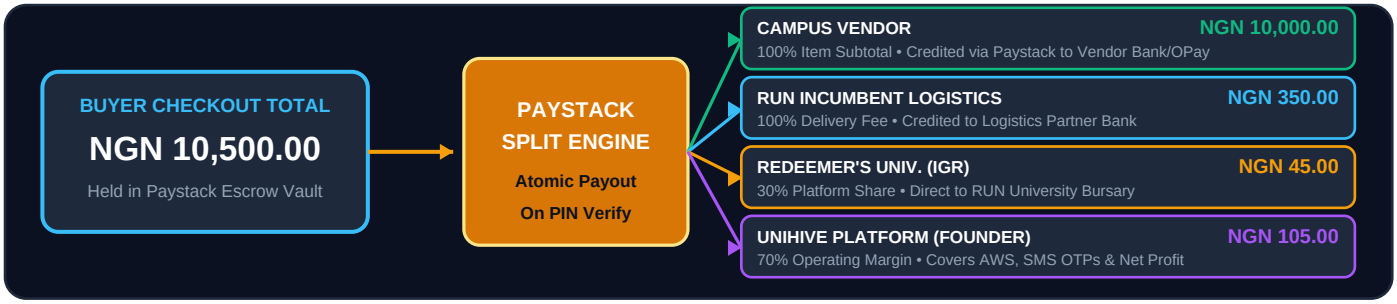
Unit Economics Benchmark: Standard NGN 10,000 Campus Transaction

Transaction Component	Amount	Recipient Entity	Operational Function & Justification
Item Subtotal	**NGN 10,000.00**	Campus Vendor	100% of product sale credited to vendor's OPay/Bank account upon PIN entry.
Campus Delivery Fee	**NGN 350.00**	RUN Incumbent Logistics Co.	100% to established campus logistics partner for handling pickup and porter lodge dropoff.
Platform Tech Fee	**NGN 150.00**	Split: UniHive & School	Covers Paystack payment charges, cloud servers, SMS OTP alerts, and net profit margin.
TOTAL PAID BY BUYER	**NGN 10,500.00**	**Paystack Escrow Vault**	**100% locked in escrow until physical inspection at porter lodge desk.**

The Platform Tech Fee Split (NGN 150 per order):

• **30% to Redeemer's University Bursary (NGN 45 per order):** Direct institutional royalty remitted to the University Bursary, Student Welfare Fund, or Directorate of Entrepreneurship.
• **70% to UniHive Technologies / Founder (NGN 105 per order):** Covers third-party cloud infrastructure (AWS/PostgreSQL), SMS OTP alerts (Termii), and generates founder net operating profit.
• **Semester Vendor Accreditation Fee:** NGN 2,000 per verified student merchant per semester, split 50/50 (NGN 1,000 to RUN Entrepreneurship Development Center, NGN 1,000 to UniHive for seller verification badges).

Figure 4: Automated Financial Settlement & Revenue Split Architecture



Monthly & Annual Revenue Projections for Redeemer's University (RUN Flagship)

Based on Redeemer's University resident population of ~8,500 students and 1,200 faculty/staff, conservative, moderate, and mature campus adoption models project the following financials:

Adoption Scenario	Monthly Orders	Est. Monthly GMV	Incumbent Logistics Earnings / Mo	Founder Net Profit / Mo	RUN School IGR / Month	RUN School IGR / Year (ARR)
Phase 1: Conservative (15% Student Adoption)	**2,500**	NGN 15.0M	NGN 875,000	**NGN 185,000**	**NGN 112,500**	**NGN 1,350,000**
Phase 2: Moderate Target (35% Student Adoption)	**6,000**	NGN 38.0M	NGN 2,100,000	**NGN 480,000**	**NGN 270,000**	**NGN 3,240,000**
Phase 3: Mature Campus (60% Full Engagement)	**12,000**	NGN 80.0M	NGN 4,200,000	**NGN 1,020,000**	**NGN 540,000**	**NGN 6,480,000**

7. Omnichannel UI/UX Architecture & Interface Showcase

UniHive is fully designed, prototyped, and built with responsive fidelity. It features a dual-theme design system: **Clean Campus Light Mode** for students and merchants across mobile and desktop, and a high-focus **Operations Dark Mode** exclusively for the Administrative Command Center.

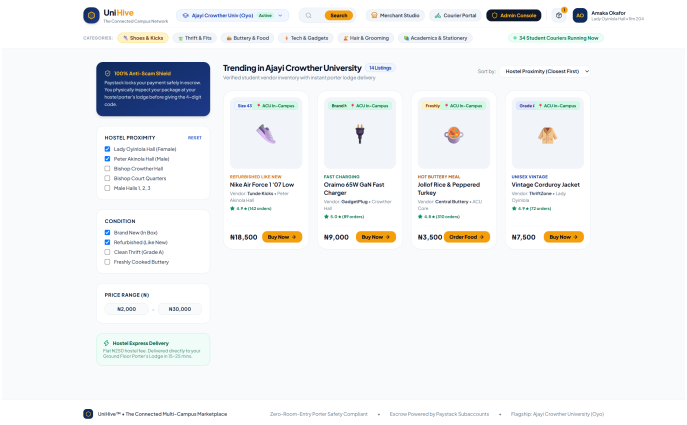


Figure 5A: Desktop Marketplace Hub with RUN Hostel Proximity Filter

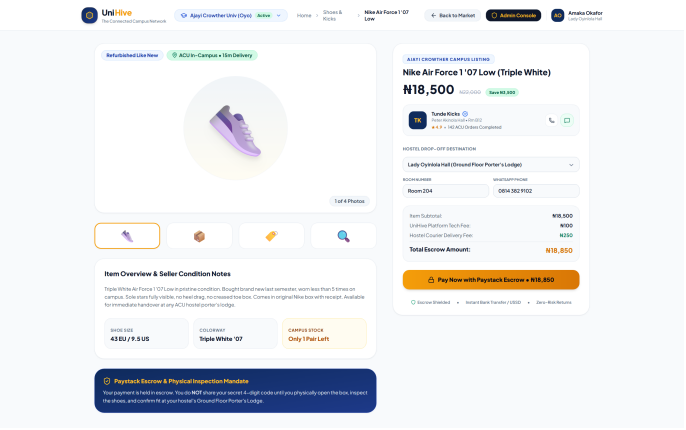
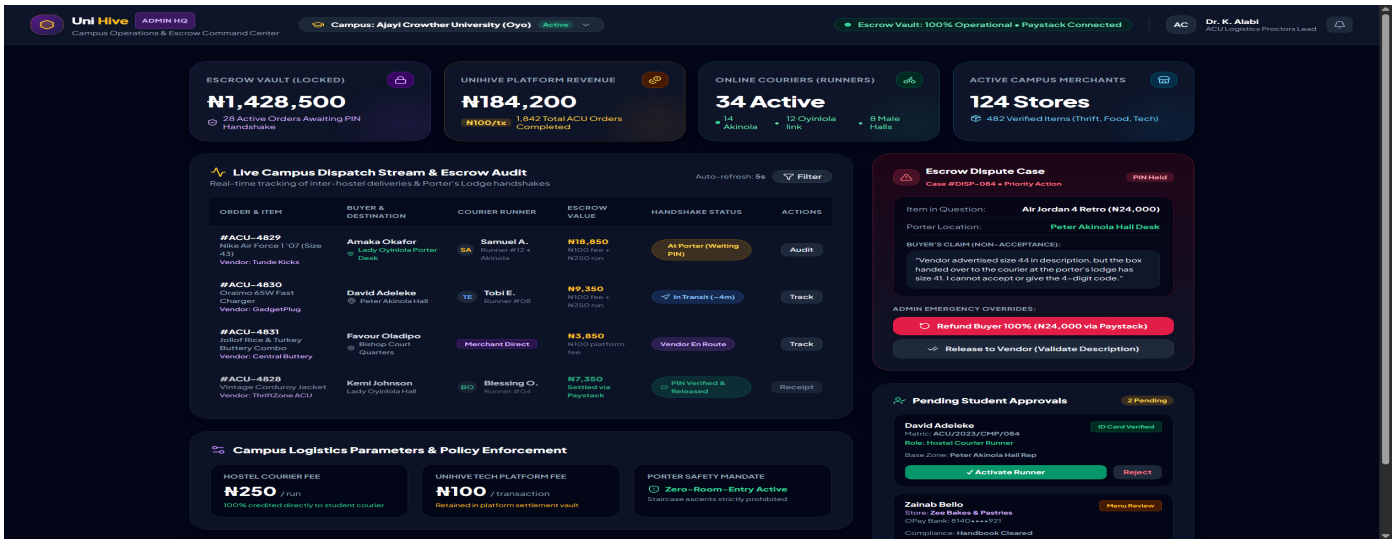


Figure 5B: Split-Column Product Checkout with Escrow Audit

Figure 6: UniHive™ Admin Operations Command Center (Dark Mode with Real-Time Escrow Vault & Proctor Stream)



Codebase & Automated Verification Status

The underlying backend engine is built with Node.js, TypeScript, Express, and Prisma ORM, verified across **25/25 automated integration tests (100% pass rate)**. Verified features include: exact escrow fee calculation, 4-digit bcrypt PIN verification, 3-attempt brute-force lockouts, atomic Paystack subaccount settlements, and strict `@run.edu.ng` domain authorization.

8. Implementation Roadmap & Stakeholder Rollout Plan

To ensure a seamless, disciplined launch at Redeemer's University without disrupting ongoing academic or hostel activities, UniHive proposes an 8-week structured rollout across three phased milestones:

Phase & Timeline	Key Operational Milestones	Involved University Stakeholders
PHASE 1: WEEKS 1 - 2 Institutional Alignment & Partner Onboarding	• Presentation to Faculty Advisor & Directorate of Student Affairs. • Formal introductory meeting with RUN's incumbent campus logistics service provider. • API sandbox configuration and logistics dispatcher training. • Security Proctorate briefing on ground-floor porter lodge protocol.	• Faculty Advisor • Dean, Student Affairs • Campus Logistics Partner • Chief Proctor
PHASE 2: WEEKS 3 - 5 Controlled Closed Pilot (Daniel & Queen Esther)	• Onboard 20 verified student merchants across fashion, tech, and study notes. • Restrict delivery routes to Daniel Hall (Male) and Queen Esther Hall (Female). • Monitor 250 initial escrow orders, porter lodge handshakes, and PIN entries. • Collect weekly satisfaction feedback from hall wardens and porters.	• Hall Wardens (Daniel & Queen Esther) • Logistics Dispatch Team • Verified Pilot Vendors • Student Buyers
PHASE 3: WEEKS 6 - 8 Full Campus Expansion & Bursary Remittance	• Expand delivery coverage to Deborah, Dorcas, Joseph, Samuel, and staff quarters. • Activate University Bursary automated 25%–30% IGR remittance account. • Open public merchant applications through RUCED. • Present post-launch performance evaluation to university management.	• University Bursary • Director, RUCED • All Hall Porters • General Student Body

9. Prayers & Actions Respectfully Requested from Faculty Advisor

To advance this homegrown student technological innovation into an officially sanctioned campus project, the student founder respectfully requests the following mentorship and administrative support from our lecturer/advisor:

- **1. Faculty Academic Mentorship & Sponsorship:** Acceptance of this project under faculty advisory oversight as a model university-industry digital transformation initiative originating from the Department of Computer Science.
- **2. Official Recommendation to Student Affairs & RUCED:** A formal letter of introduction to the Dean of Student Affairs and the Director of the Redeemer's University Centre for Entrepreneurship Development (RUCED) to present this institutional proposal.
- **3. Facilitation of Strategic Logistics Dialogue:** Administrative assistance in scheduling an introductory technical alignment session with the leadership of Redeemer's University's incumbent campus logistics company to formalize delivery protocol integration.
- **4. Pilot Approval for Daniel & Queen Esther Halls:** Support in obtaining administrative consent from the Hall Wardens and Student Affairs Directorate for the 3-week closed beta pilot in Daniel and Queen Esther Halls.

RESPECTFULLY SUBMITTED BY:

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Student Founder & Lead Software Architect
Department of Computer Science
Redeemer's University (RUN), Ede, Osun State
Email: founder@run.edu.ng | Tel: +234 (0) 812 345 0001

FACULTY ADVISOR ACKNOWLEDGEMENT:

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Faculty Academic Advisor / Senior Lecturer
Department of Computer Science / Directorate of Student Affairs
Redeemer's University (RUN), Ede, Osun State
Date: ____ / ____ / 2026 | Signature: _____